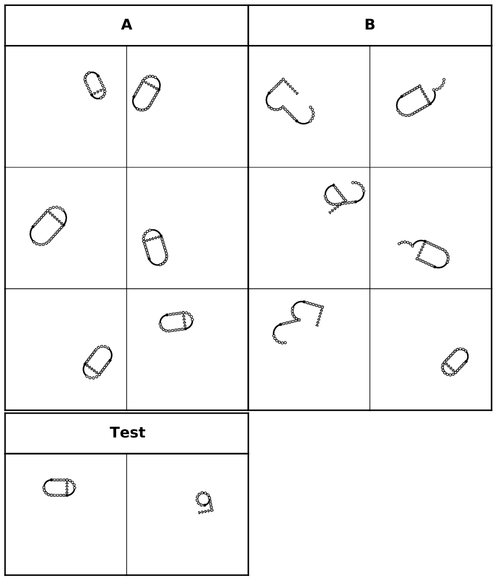
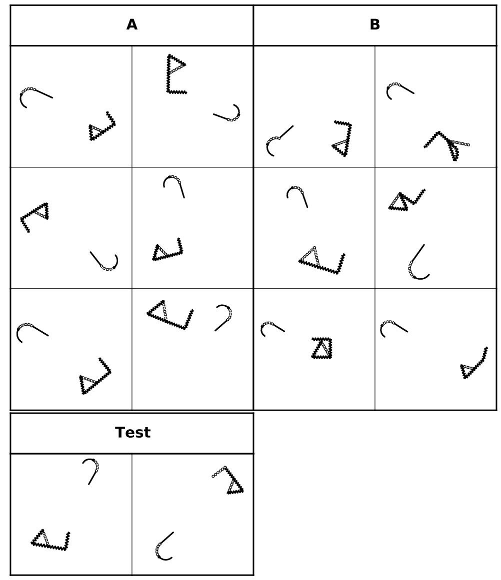
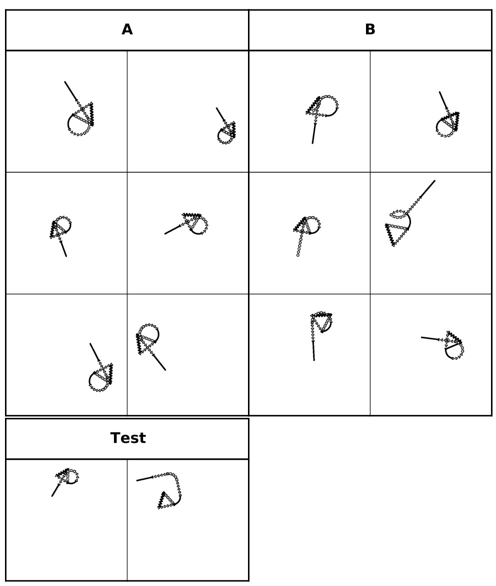
D.1 More Examples of Free-Form Shape Problems



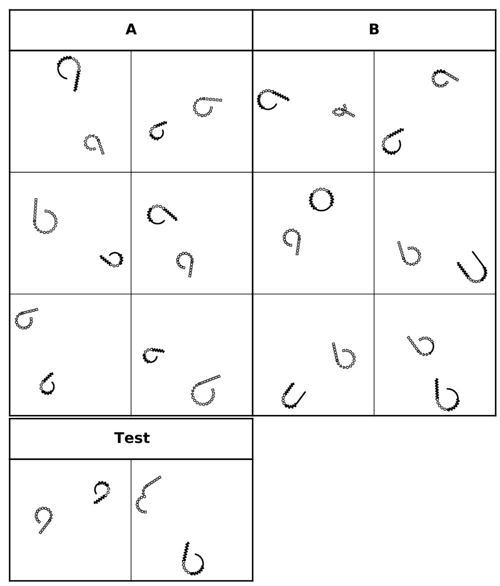
(a) # of actions: [7]



(b) # of actions: [3, 5]



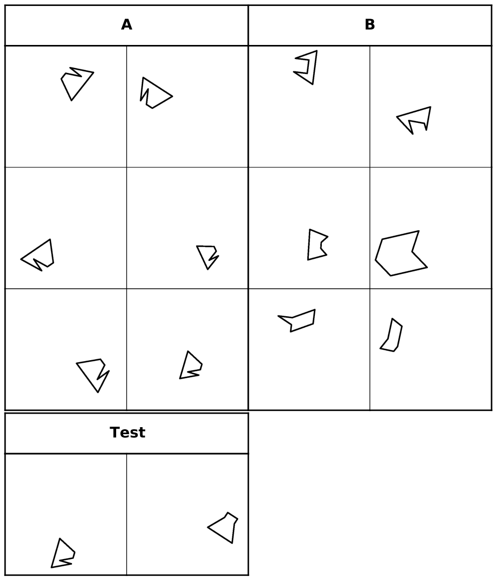
(c) # of actions: [8]



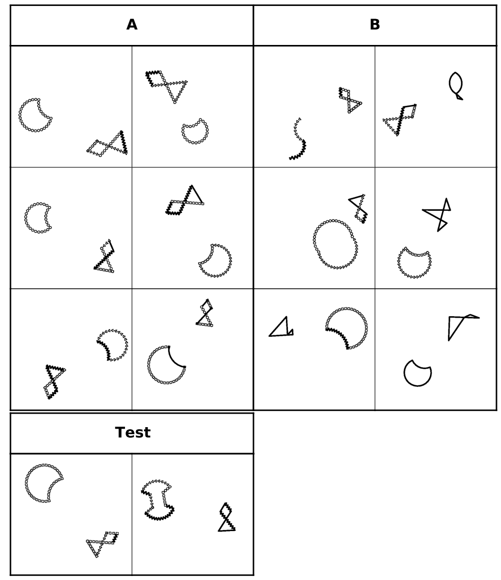
(d) # of actions: [4, 4]

Figure 7: More examples of free-form shape problems, where (a) the shape is generated by five action strokes, (b) the two shapes are generated by three and five action strokes, respectively, (c) the shape is generated by eight action strokes, (d) the two shapes are generated by four and four action strokes, respectively. In each problem, set $\mathcal{A}$ contains six images that satisfy the concept and set $\mathcal{B}$ contains six images that violate the concept. We also show two test images (left: positive, right: negative) in the binary classification problems. In free-form shape problems, the task is about discovering the concept of base strokes in a shape, which may differ in inter-stroke angles or stroke types (i.e., normal lines, zigzag lines, normal arcs, arcs formed by a set of circles, etc.). As we can see from these examples, the difference in stoke types may be subtle to distinguish. We do not distinguish concepts by the shape size and orientation, absolute position, or relative distance of two shapes.

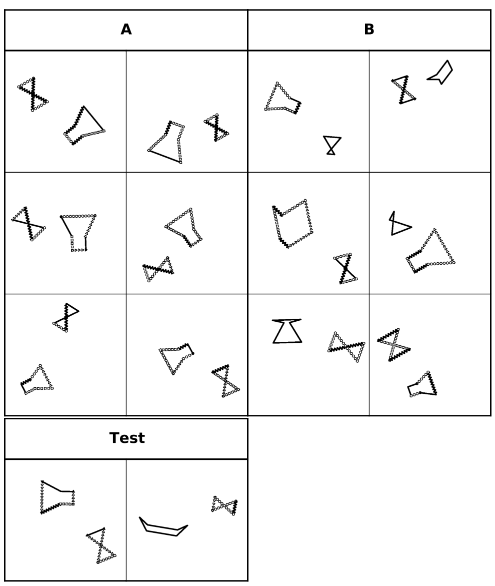
D.2 More Examples of Basic Shape Problems



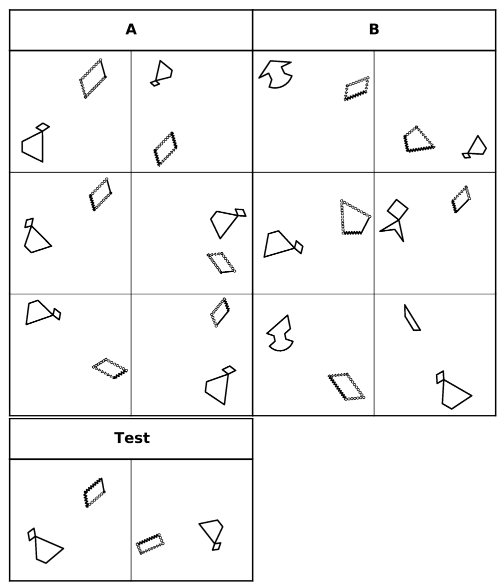
(a) `mountains`



(b) `nearly_full_moon` and `fish`



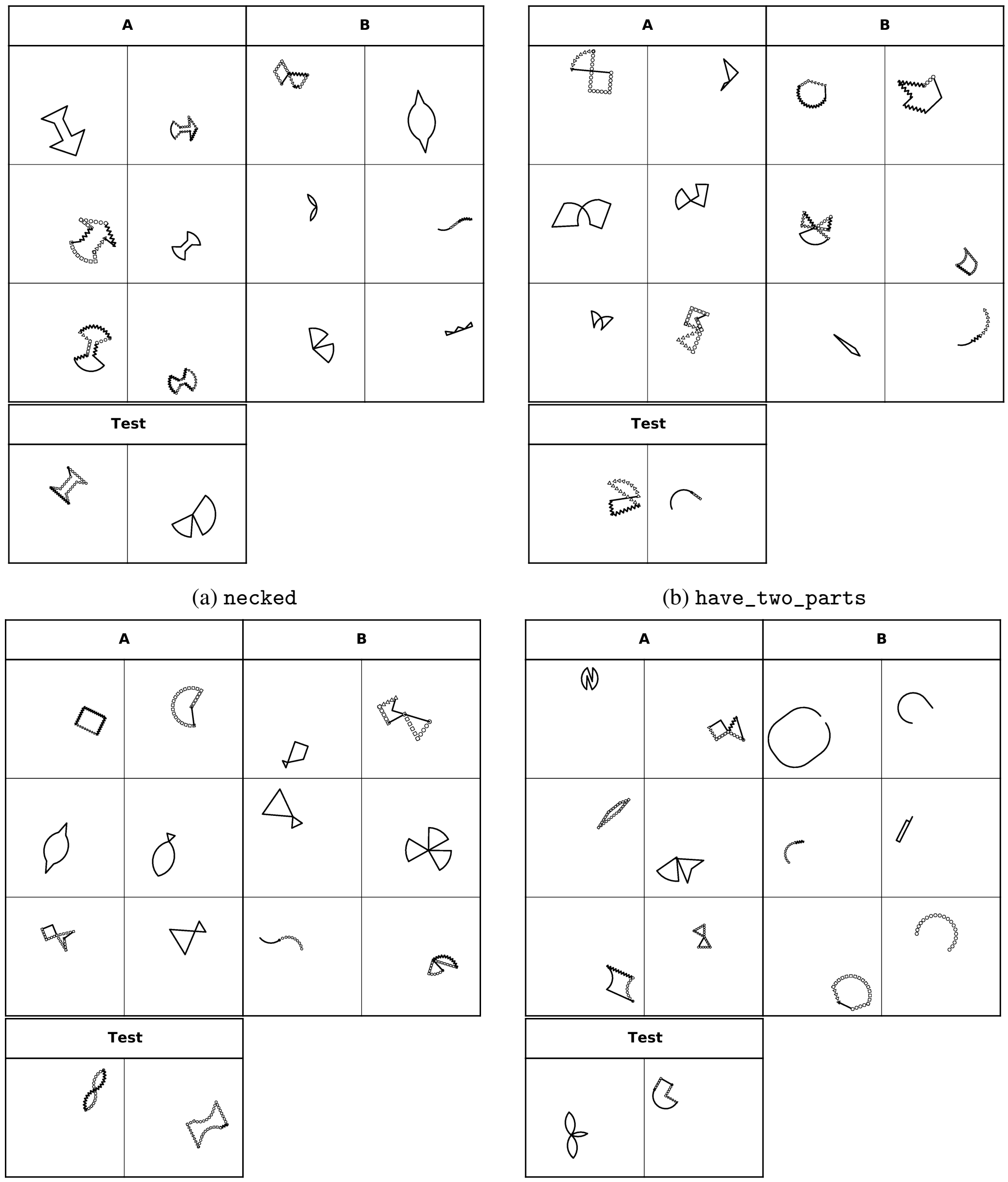
(c) `funnel` and `hourglass`



(d) `turtle` and `parallelogram`

Figure 8: More examples from basic shape problems, where (a) the concept is `mountains`, (b) the concept is a combination of `nearly_full_moon` and `fish`, (c) the concept is a combination of `funnel` and `hourglass`, (d) the concept is a combination of `turtle` and `parallelogram`. In each problem, set $\mathcal{A}$ contains six images that satisfy the concept and set $\mathcal{B}$ contains six images that violate the concept. We also show two test images (left: positive, right: negative) in the binary classification problems. In basic shape problems, the task is about discovering the concept of the shape category itself. We do not distinguish concepts by the shape size and orientation, absolute position, or relative distance of two shapes.

D.3 More Examples of Abstract Shape Problems



(a) `necked` (b) `have_two_parts`

(c) `symmetric` (d) `have_acute_angle`

Figure 9: More examples of abstract shape problems, where one attribute shared by the positive images in set $\mathcal{A}$ is considered as the underlying concept. In particular, (a) the concept is `necked`, (b) the concept is `have_two_parts` (it means the shape can be separated into two disconnected parts by a connecting point), (c) the concept is `symmetric`, (d) the concept is `have_acute_angle`. In each problem, set $\mathcal{A}$ contains six images that satisfy the concept and set $\mathcal{B}$ contains six images that violate the concept. We also show two test images (left: positive, right: negative) in the binary classification problems. We do not distinguish concepts by the shape size and orientation, absolute position, or relative distance of two shapes. We can see for abstract shape problems, it may also be challenging for humans without clearly understanding the meanings of abstract attributes.